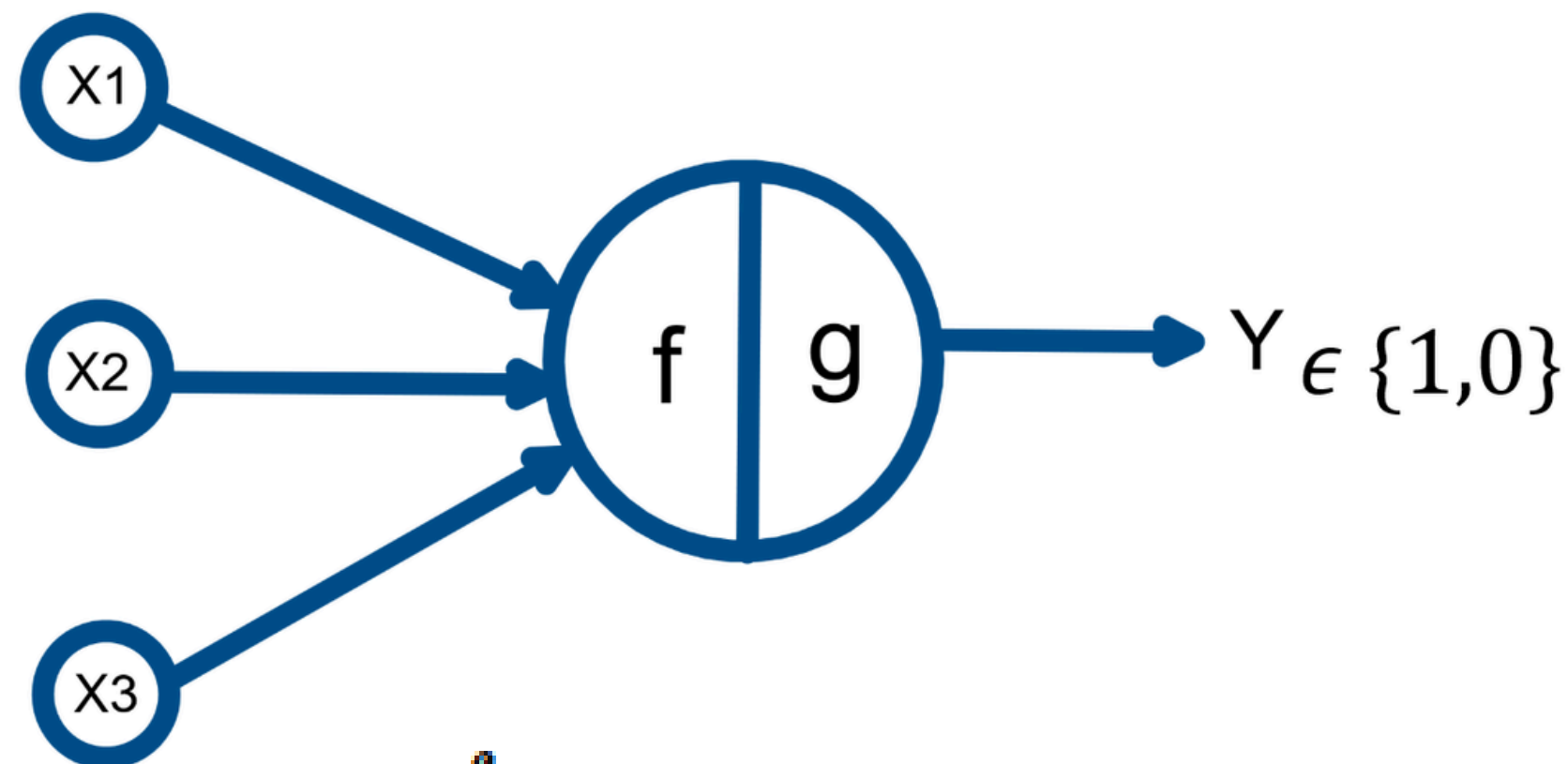




McCulloch Pitts Neuron

$$x_1, x_2, x_3 \in \{0, 1\}$$

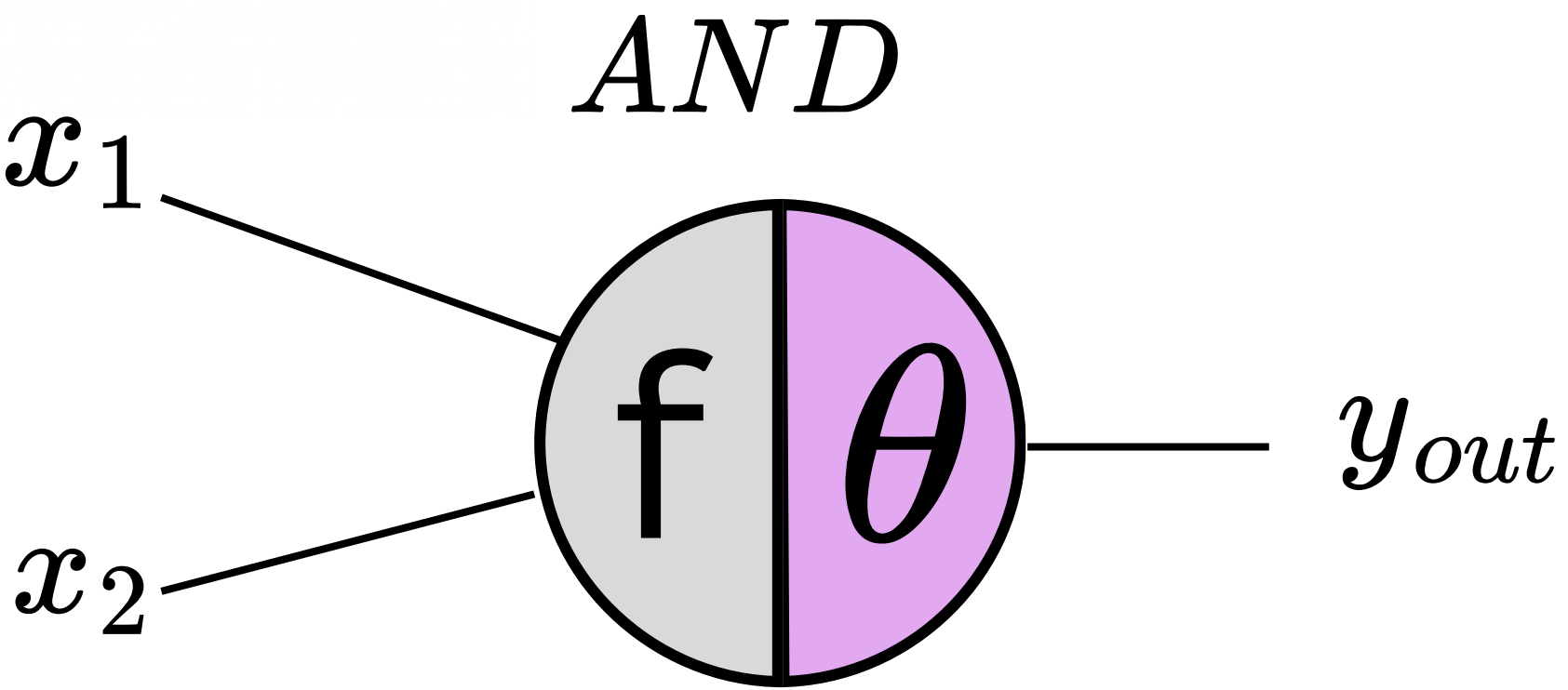


$$f = x_1 + x_2 + x_3$$

$$g(f) = \left\{ \begin{array}{ll} 1, & f \geq T \\ 0, & f < T \end{array} \right\} \text{Threshold logic}$$

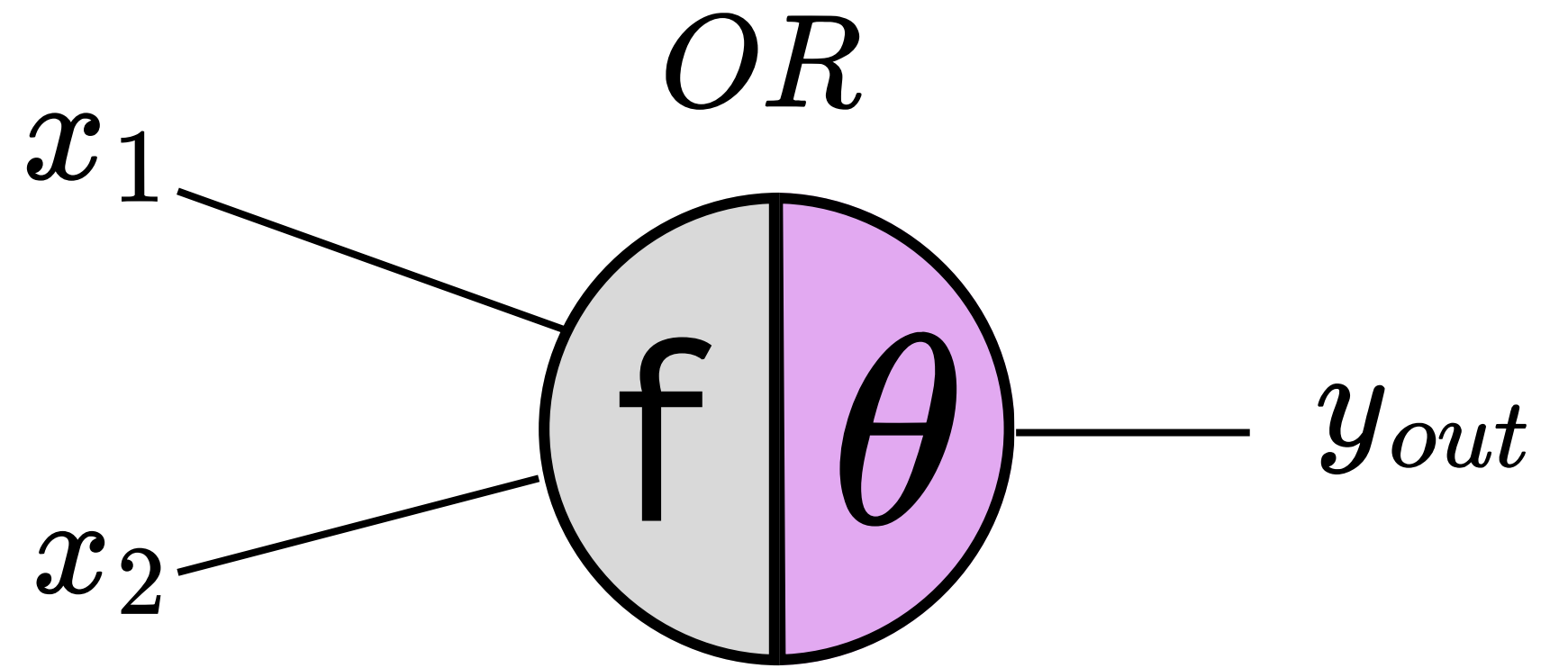


Logic Gates With MCP neuron



$$\theta \geq 2$$

x_1	x_2	$x_1 \wedge x_2$
0	0	0
0	1	0
1	0	0
1	1	1



$$\theta \geq 1$$

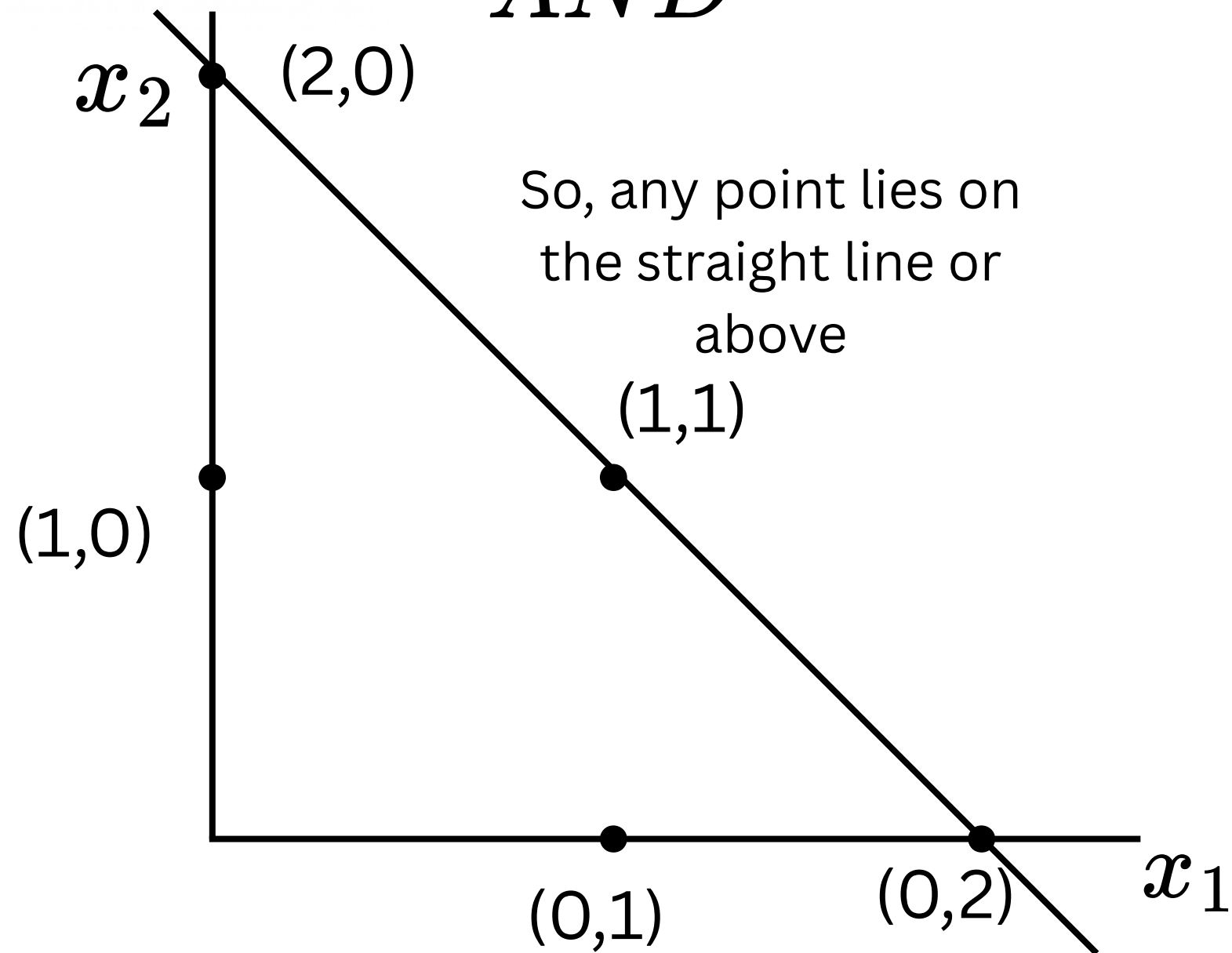
x_1	x_2	$x_1 \vee x_2$
0	0	0
0	1	1
1	0	1
1	1	1

$$x_i \in \{0, 1\}$$



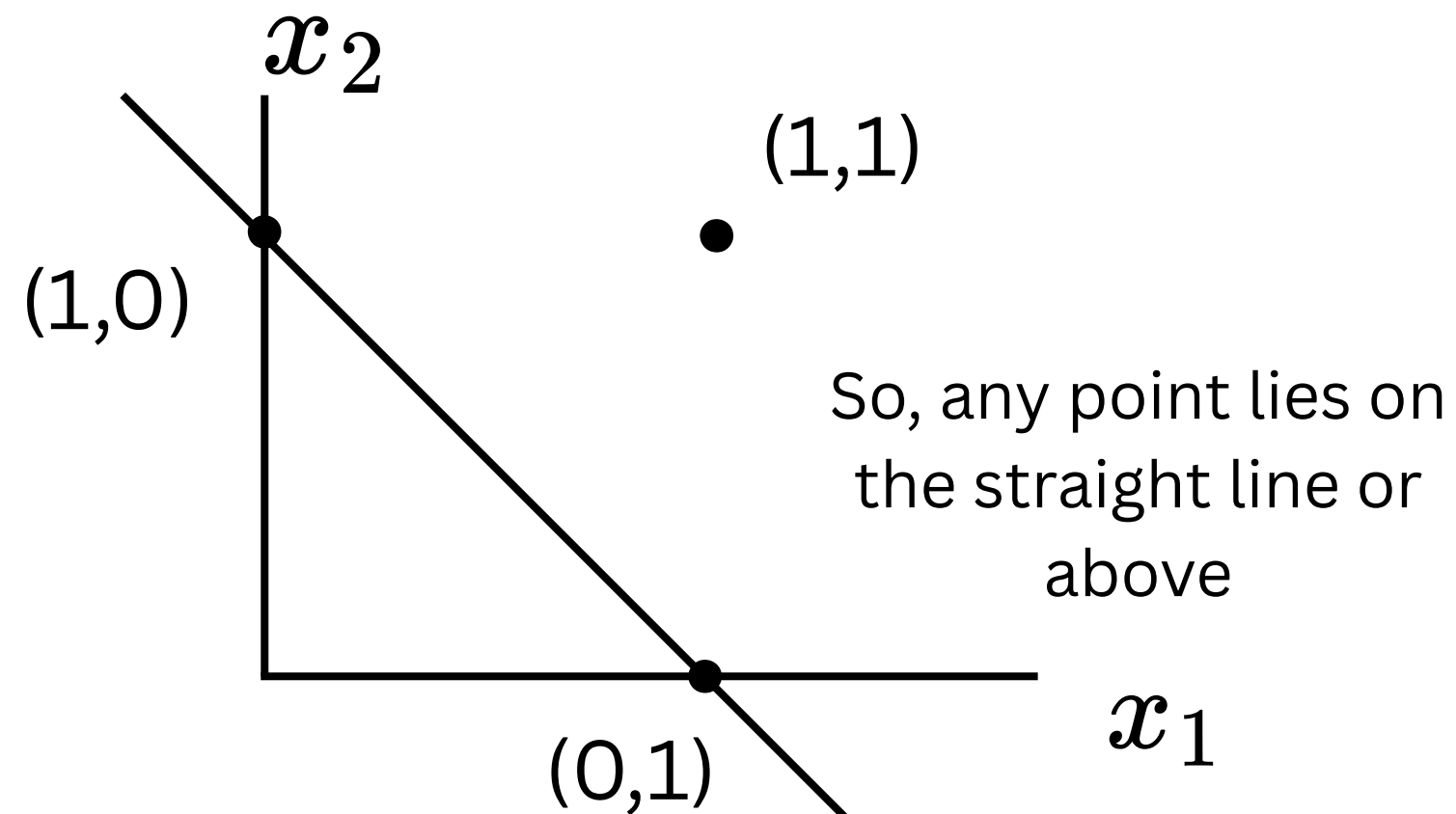
Logic Gates With MCP neuron

AND



$$\theta \geq 2$$

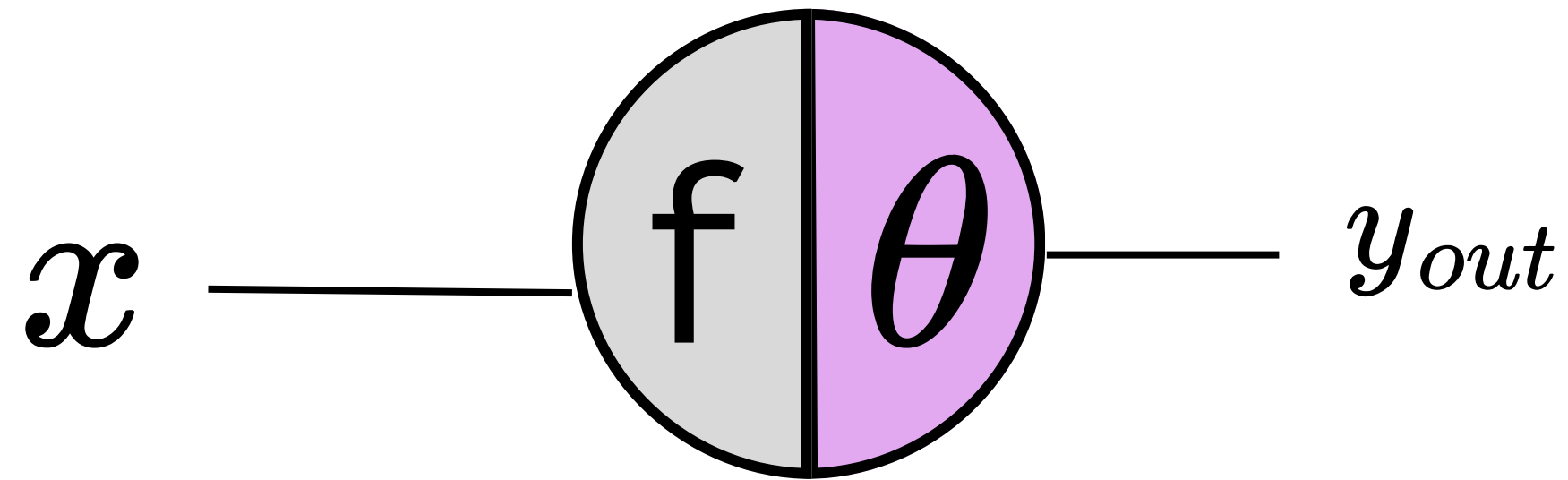
OR



$$\theta \geq 1$$



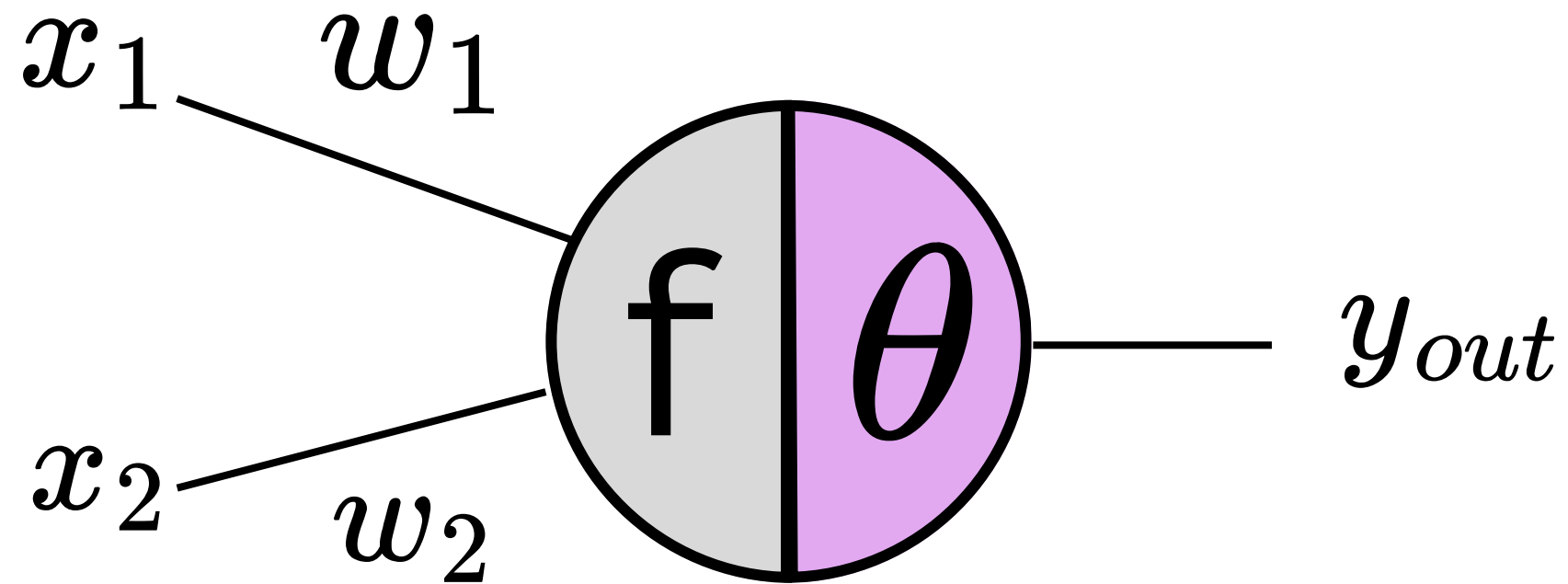
Logic Gates With MCP neuron



$$x \in \{0, 1\}$$

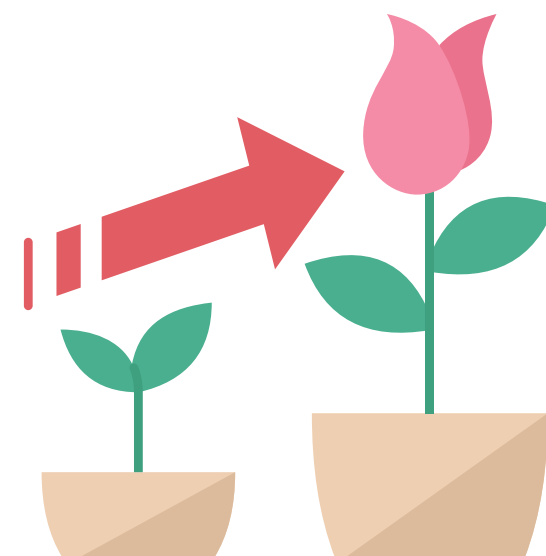


Perceptron



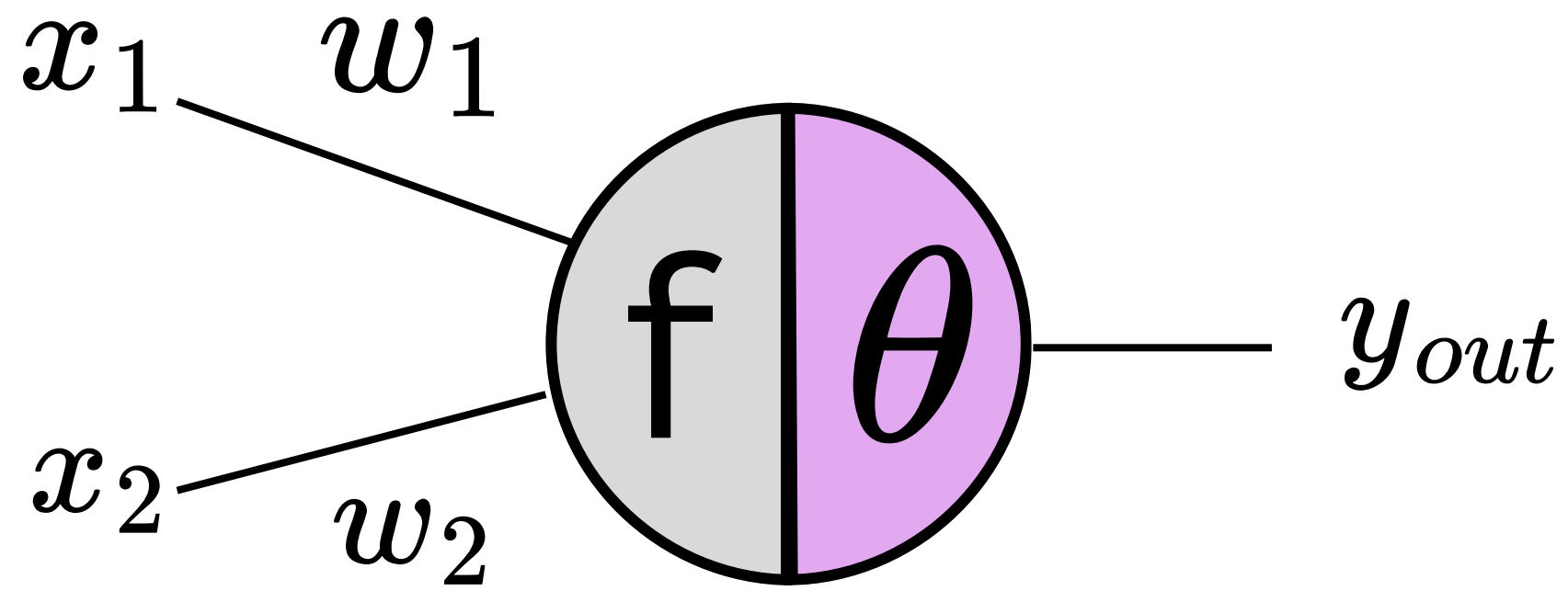
$$f = \sum_{i=1}^2 w_i x_i$$

- It is the same idea followed by MCP neuron architecture except the introduction of weights
- The weight gives power to keep decision boundary at any place

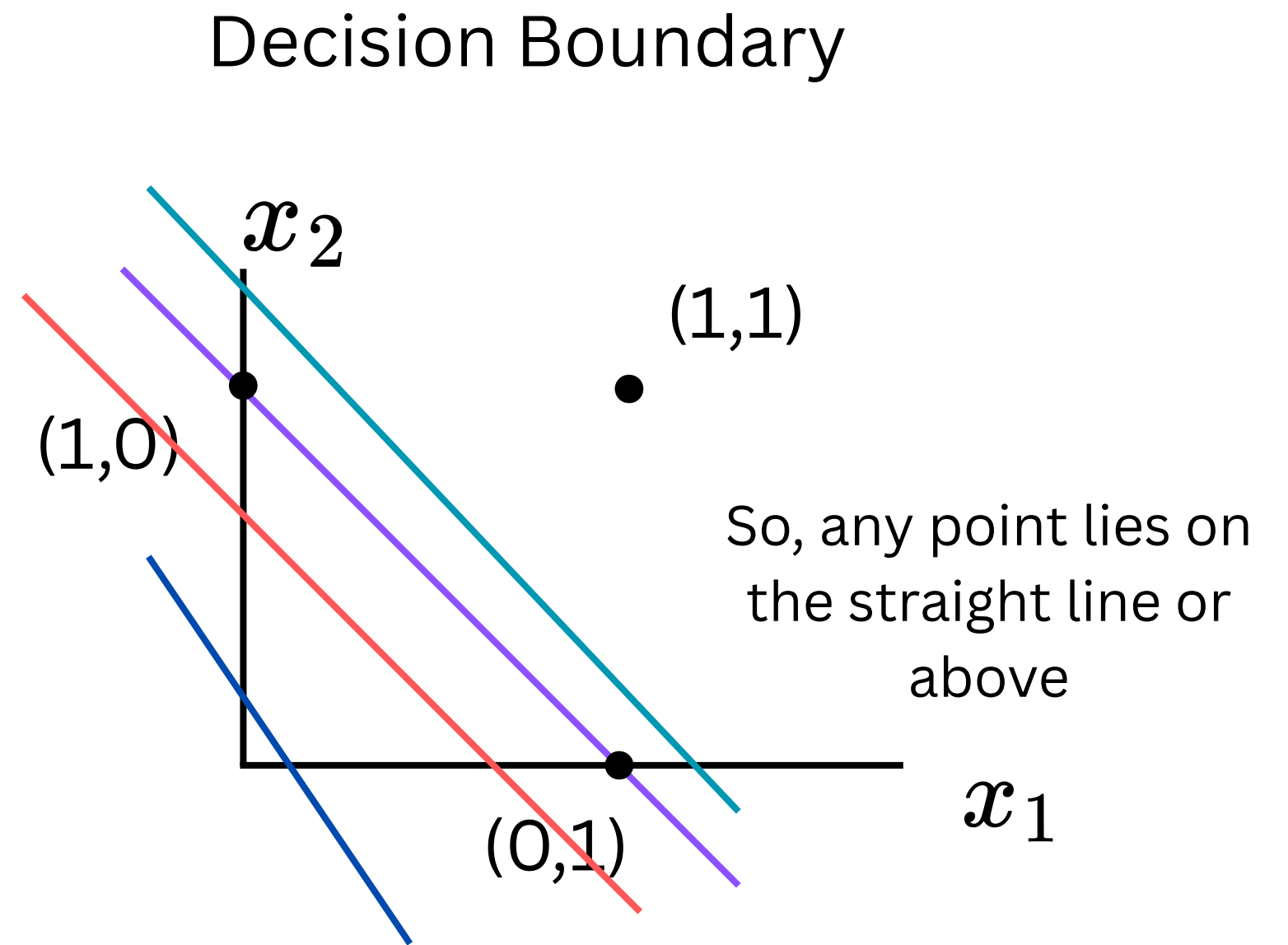




Perceptron

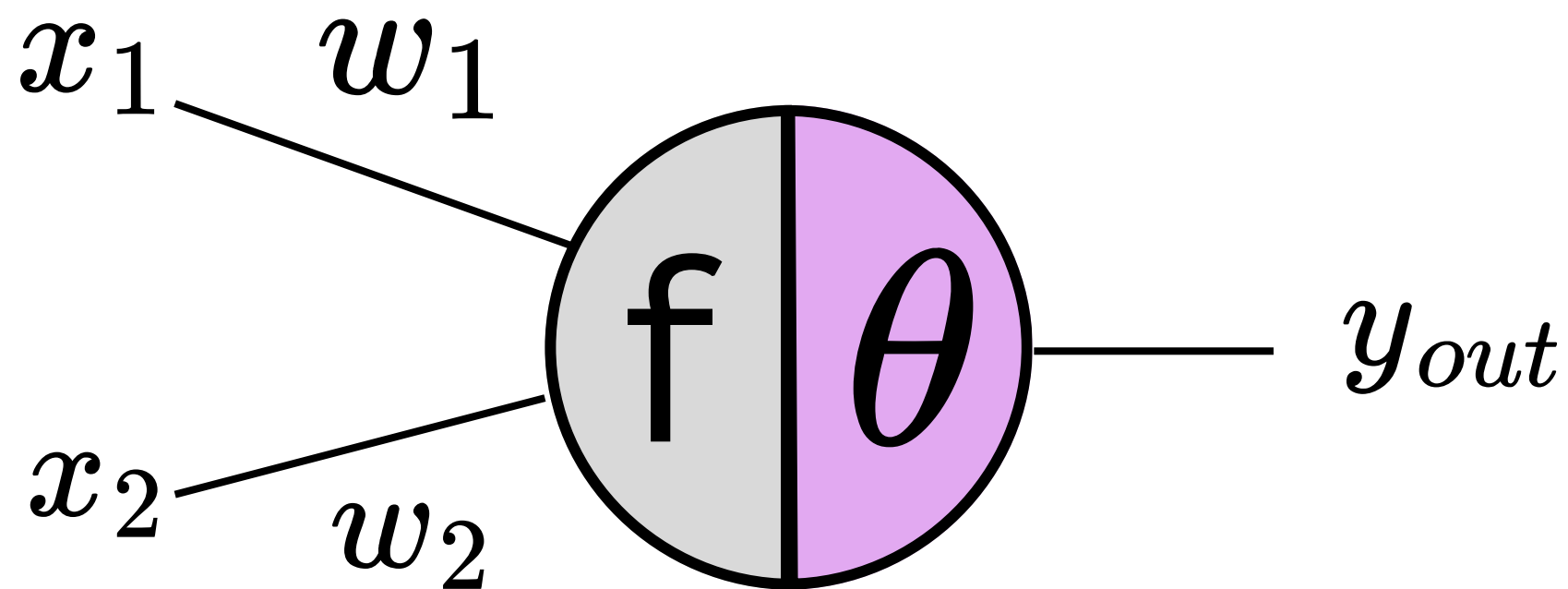


$$f = \sum_1^2 w_i x_i$$





Perceptron



$$f = \sum_1^2 w_i x_i$$

Rewrite

$$\sum_{i=1}^n w_i x_i - \theta \geq 0$$

Define

$$w_0 = -\theta$$

$$x_0 = 1$$

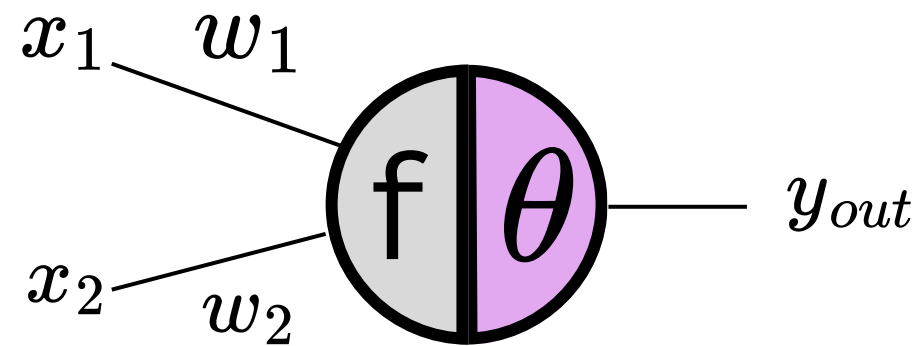
Then

$$w_0 x_0 + \sum_{i=1}^n w_i x_i \geq 0$$

or



Perceptron



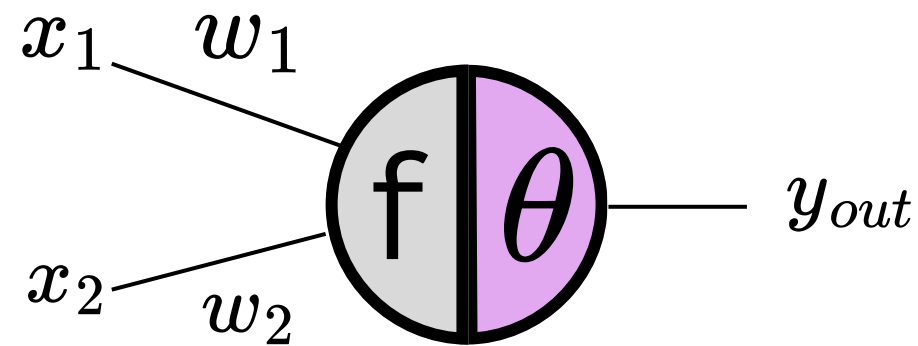
$$f = \sum_{i=1}^2 w_i x_i$$

OR Truth Table with Perceptron Conditions

x_1	x_2	OR	Perceptron Condition
0	0	0	$w_0 + \sum_{i=1}^2 w_i x_i < 0$
0	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
1	0	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
1	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$



Perceptron



$$f = \sum_1^2 w_i x_i$$

Substituting Individual Inputs

For (0, 0)

$$w_0 + w_1(0) + w_2(0) < 0$$

$$w_0 < 0$$

For (0, 1)

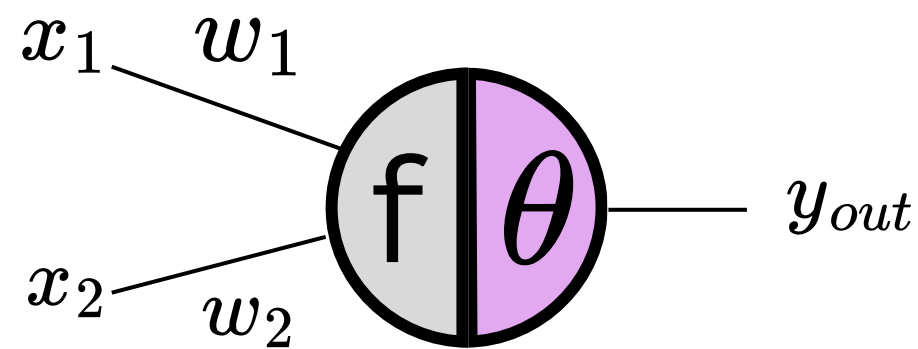
$$w_0 + w_1(0) + w_2(1) \geq 0$$

$$w_0 + w_2 \geq 0$$

$$w_2 \geq -w_0$$



Perceptron



$$f = \sum_1^2 w_i x_i$$

For (1, 0)

$$w_0 + w_1(1) + w_2(0) \geq 0$$

$$w_0 + w_1 \geq 0$$

$$w_1 \geq -w_0$$

For (1, 1)

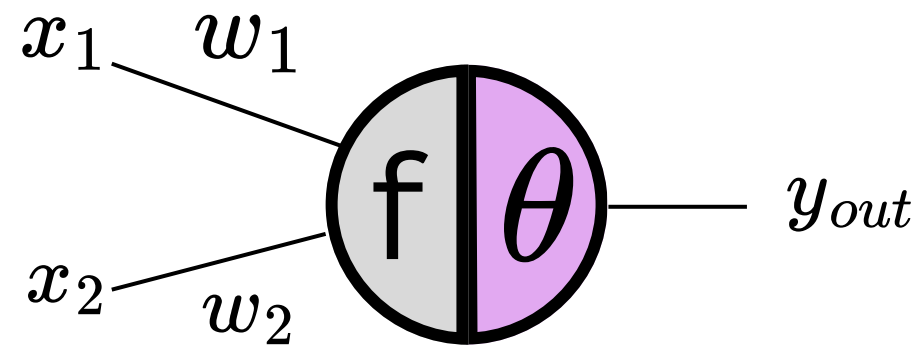
$$w_0 + w_1(1) + w_2(1) \geq 0$$

$$w_0 + w_1 + w_2 \geq 0$$

$$w_1 + w_2 \geq -w_0$$



Perceptron



$$f = \sum_1^2 w_i x_i$$

Final Constraints

$$w_0 < 0$$

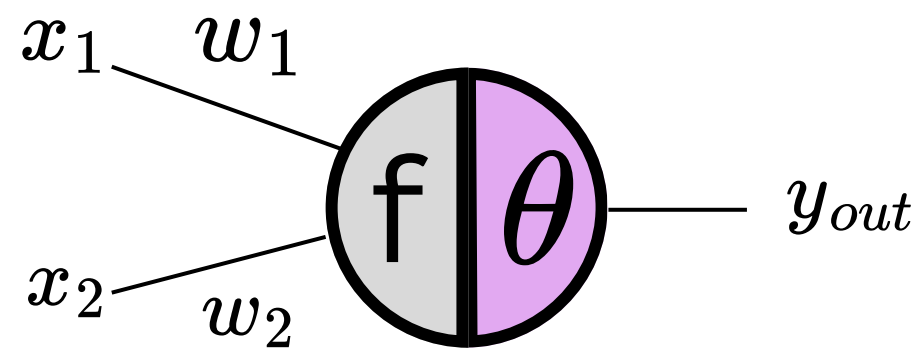
$$w_1 \geq -w_0$$

$$w_2 \geq -w_0$$

$$w_1 + w_2 \geq -w_0$$



Perceptron



$$f = \sum_1^2 w_i x_i$$

One Possible Solution

$$w_0 = -1$$

$$w_1 = 1.1$$

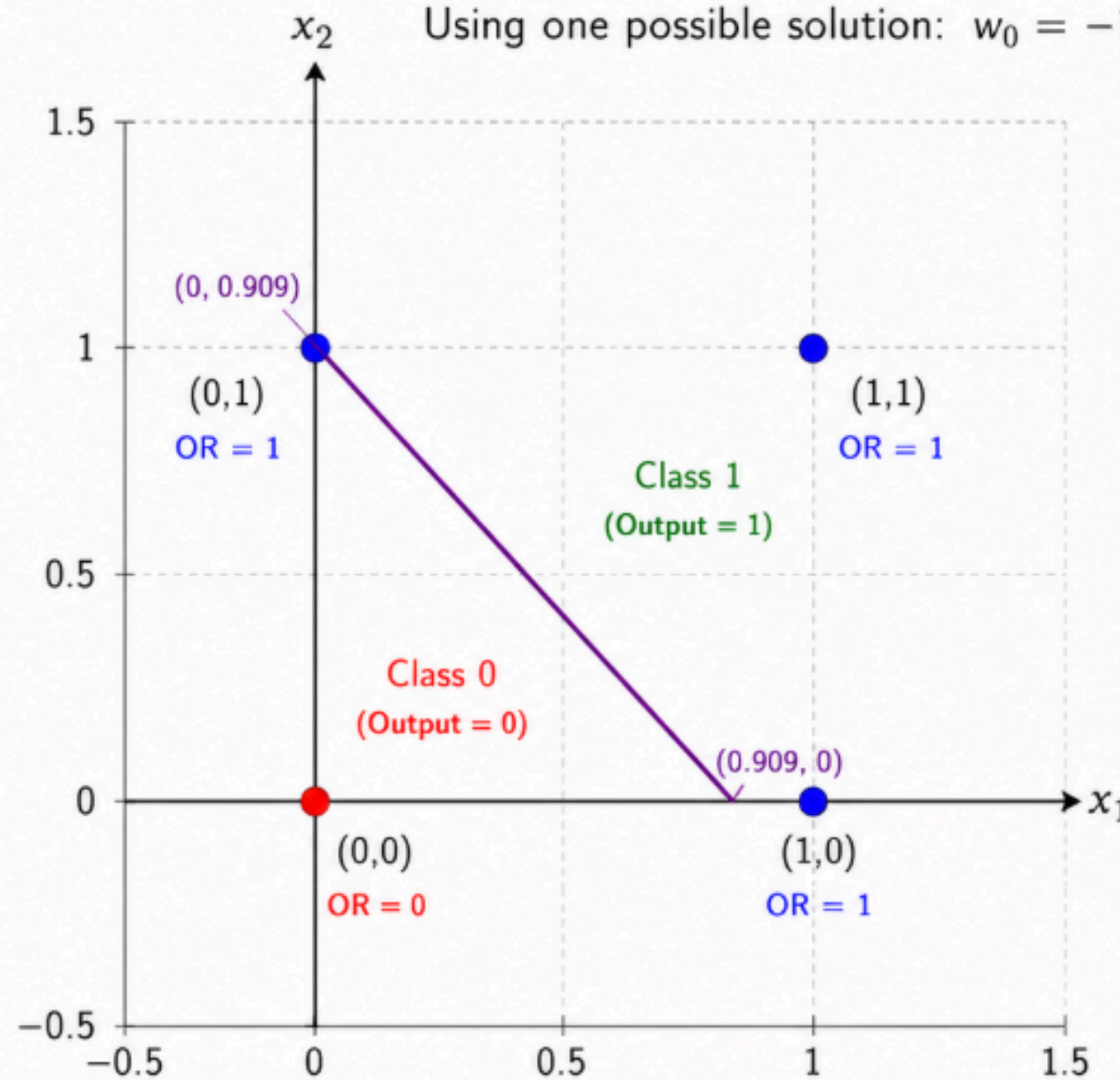
$$w_2 = 1.1$$



Perceptron

Decision Boundary for OR Gate Using Perceptron

Using one possible solution: $w_0 = -1$, $w_1 = 1.1$, $w_2 = 1.1$



Decision Boundary Equation

$$\begin{aligned}w_0 + w_1x_1 + w_2x_2 &= 0 \\-1 + 1.1x_1 + 1.1x_2 &= 0 \\x_2 &= -x_1 + \frac{1}{1.1} = -x_1 + 0.909\end{aligned}$$

Intercepts

- When $x_1 = 0 \Rightarrow x_2 = 0.909 \Rightarrow (0, 0.909)$
- When $x_2 = 0 \Rightarrow x_1 = 0.909 \Rightarrow (0.909, 0)$

Classification Rule

$$\begin{aligned}w_0 + w_1x_1 + w_2x_2 \geq 0 &\Rightarrow \text{Class 1} \\w_0 + w_1x_1 + w_2x_2 < 0 &\Rightarrow \text{Class 0}\end{aligned}$$

- $\bullet$ OR = 1
- $\bullet$ OR = 0
- — Decision Boundary

Verification with the chosen weights

$$\begin{aligned}(0,0): \\-1 + 0 + 0 &= -1 < 0 \\ \Rightarrow \text{OR} = 0 \quad \checkmark\end{aligned}$$

$$\begin{aligned}(0,1): \\-1 + 0 + 1.1 &= 0.1 \geq 0 \\ \Rightarrow \text{OR} = 1 \quad \checkmark\end{aligned}$$

$$\begin{aligned}(1,0): \\-1 + 1.1 + 0 &= 0.1 \geq 0 \\ \Rightarrow \text{OR} = 1 \quad \checkmark\end{aligned}$$

$$\begin{aligned}(1,1): \\-1 + 1.1 + 1.1 &= 1.2 \geq 0 \\ \Rightarrow \text{OR} = 1 \quad \checkmark\end{aligned}$$



Perceptron

Perceptron Decision Boundary for OR Gate

OR Truth Table		
x_1	x_2	OR (Desired Output)
0	0	0
0	1	1
1	0	1
1	1	1

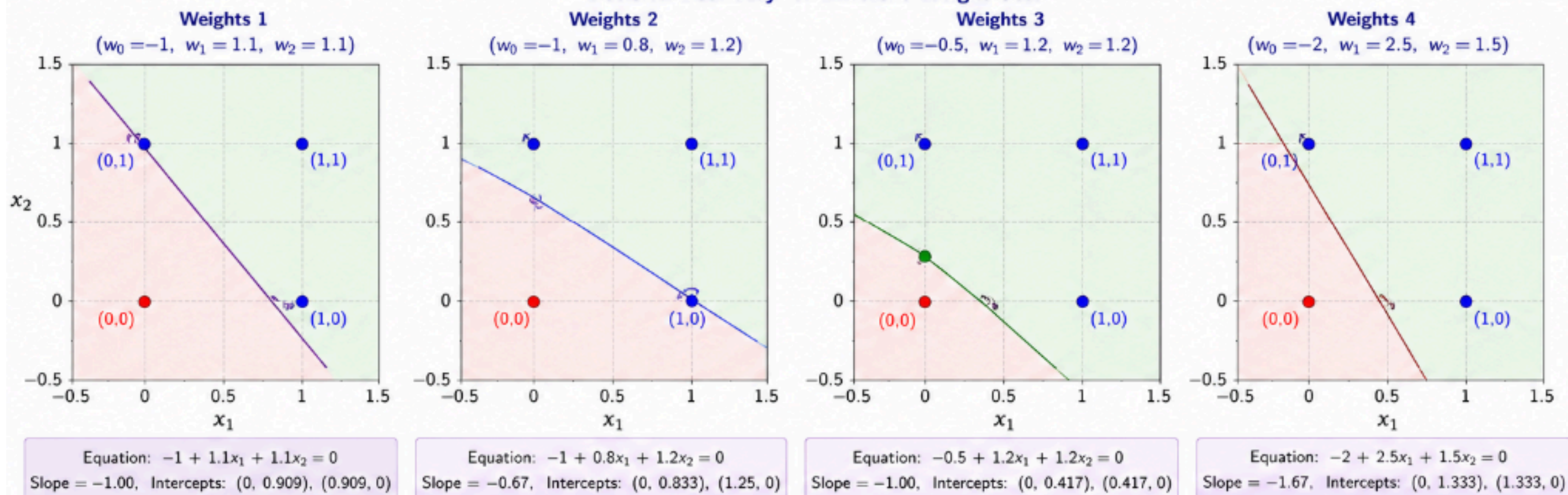
Perceptron Model

- Output $y = 1$, if $w_0 + w_1x_1 + w_2x_2 \geq 0$
- Output $y = 0$, if $w_0 + w_1x_1 + w_2x_2 < 0$

Decision Boundary

$$w_0 + w_1x_1 + w_2x_2 = 0 \Rightarrow x_2 = -\frac{w_1}{w_2}x_1 - \frac{w_0}{w_2}$$

Decision Boundary for Different Weight Sets



Legend

- Class 1 (OR = 1) (Green background)
- Class 0 (OR = 0) (Red background)
- Training points with OR = 1 (Blue dots)
- Training point with OR = 0 (Red dot)
- Decision Boundary (Purple line)

One Possible Solution Verification (Weights 1)

(x_1, x_2)	Net Input $w_0 + w_1x_1 + w_2x_2$	Output y	OR (Desired)
(0,0)	$-1 + 1.1(0) + 1.1(0) = -1.00$	0	0 ✓
(0,1)	$-1 + 1.1(0) + 1.1(1) = 0.10$	1	1 ✓
(1,0)	$-1 + 1.1(1) + 1.1(0) = 0.10$	1	1 ✓
(1,1)	$-1 + 1.1(1) + 1.1(1) = 1.20$	1	1 ✓



Perceptron

- Each point on the surface represents a unique combination of weights (w_1, w_2).
- The height of the surface indicates the classification error.
- Higher regions correspond to poor weight choices and incorrect classifications.
- Lower regions correspond to better-performing weight combinations.
- Zero-error regions indicate weights that correctly implement the OR gate.
- Multiple weight combinations achieve the same zero error, showing that the solution is not unique.
- Learning aims to move from high-error regions toward low-error regions.
- The error surface visualizes the optimization process used in training neural networks.

Perceptron Error Surface for OR Gate

